

Niloufar Moradijam

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👤 PROFILE

XR and Creative Technology Developer with hands-on experience building real-time applications across AR, VR, WebGL, and web platforms. Develops Unity-based immersive experiences alongside React and Vue web applications integrating AI-driven interactive avatars and generative pipelines. Skilled in shader programming, procedural content generation, and cross-platform debugging. Currently completing an M.Sc. in Interactive Media (Game Development track) with thesis research on LLM-assisted procedural shader generation. Background in full-stack web development with a strong foundation in C#, C++, Python, and JavaScript. Open to relocation across Europe.

🧠 SKILLS

Programming Languages

Proficient in: C++ (Modern C++14/17), C#, Python, JavaScript,
Have Experience in: Typescript, Java

Frameworks and Libraries

Proficient in: React.js, Angular, .Net, ML-Agents
Have Experience in: Flutter, JavaFX, Django, Node.js, Bootstrap, Material UI, Three.js, WebAR tools

Dev Tools

Proficient in: Unity Engine, Unreal Engine, Git, Linux (g++, CMake, gdb), Docker, AWS
Have Experience in: Matlab, Android Studio, Figma

AI and Prompt Engineering

Proficient in Using this AI Tools: Nano banana, Copilot, Runway, Seedance, HayGen, Tavus, Akool, LemonSlice

📁 PROFESSIONAL EXPERIENCE

Unity Development and Technical Art Internship, FreshFX 🔗	Jul 2025 – Present Salzburg, Austria
- Designed and implemented real-time XR applications in Unity and Unreal Engine, creating immersive user experiences for installations and research environments. - Developed a WebGL-based AR museum application in Unity featuring image tracking, video overlays, and an interactive achievement system with animated transitions. - Built React and Vue.js web applications featuring AI-driven interactive avatars (Akool, Tavus, LemonSlice) with real-time streaming, face detection, and custom UI interactions alongside the avatar experience - Promoted to a part-time position following a successful 4-month full-time internship.	
Tutoring (Teacher Assistance), FH University Upper Austria	Mar 2025 – Present Hagenberg, Austria
Mixed Reality, Object Oriented Programming, Online Multimedia, Digital Signal Processing, Digital Image/Audio Processing, Hypermedia Systems and Architecture, Algorithmic Thinking, Artificial Intelligence	
Game Development, Psiket Academy (Sharif University) 🔗	May 2024 – Sep 2024 Tehran, Iran
Designed and programmed educational mini-games using Unity, applying quantum concepts through visual gameplay to enhance learning engagement.	
Game Development, Rhino	Mar 2023 – Aug 2023 Tehran, Iran
Developing an Android game akin to TFT of League of Legends Utilizing Unity engine and C#, called Chandkhaan	
Game Development and Design Internship, Concealland Studio	Aug 2022 – Nov 2022 Tehran, Iran
- gained proficiency in 3ds Max for Mapna project simulation, designed 10 diverse buildings - acquired fundamental Unity3D skills and game programming knowledge.	
Full-Stack Development, System Group 🔗	Nov 2020 – Jul 2022 Tehran, Iran
Contributed to enterprise web systems (Angular + .NET), improving UI performance and workflow reliability.	

📁 PROJECTS

Text-to-Shader: AI-Powered Procedural 2D Asset Generation, Thesis (in-progress)

- Developed a novel pipeline converting natural language to **Unity Shader Graphs** using **Large Language Models (Gemini)** and **Signed Distance Functions (SDFs)**.
- Engineered a deterministic **C#** builder to wrap LLM-generated **HLSL** logic into valid Unity assets, ensuring **100%** compilation stability.

- Implemented an autonomous visual feedback loop using GPT-4o (VLM) to verify geometric accuracy, achieving a **90%** success rate for complex 2D shapes.
- Automated the bootstrapping of a "**Shape Knowledge Base**," generating a library of **over 200** verified procedural SDF primitives.

Pickup and Delivery RL Agent

- Developed **RL agent** for warehouse automation using PPO algorithm, achieving sequential **pickup-and-delivery** task completion in **Unity 3D** environment using MLAgents
- Implemented **curriculum learning** with progressive difficulty levels and episode variation strategy to optimize training efficiency, in a **12-dimensional** observation space
- Achieved **97% success rate** in full task after 2 Million training steps

The Holy Mountain – VR Interactive Environment , Second Semester Project

- Developed a **VR simulation of Alps** with interactive tree lifecycle, seed planting, and AI-driven animal behaviors.
- Implemented immersive VR mechanics (inventory, teleportation, stone throwing), optimized **Gaia Pro terrains**, and built custom shaders and day-night cycles. (**Unity, C#, URP, Gaia Pro, NavMesh, Shader Graph, Oculus**)

Mars Rover Drilling Simulation

- Collaborated in a **team** to develop, using **Git**, and implemented drilling system using a **custom URP shader** with terrain-aware **VFX**, arm animation, and particles — optimized for **real-time performance**.

Changing Localized Body Temperature and Thermal Perception in Virtual Reality

- Developed a **VR environment** in **Unity(C#)** and Implemented **hand tracking** and thermal perception **simulation** with Meta Quest devices.
- **Collected data** on hand positions, skin temperature, and survey responses to analyze thermal cues in VR (**Prototype**)

Interactive Sakura Tree

- Developed a **2D Sakura tree scene** using **OpenGL (C++)**, **GLFW**, and **GLAD**, as part of coursework for Real-Time Graphics.
- Implemented **dynamic transformations** including **interactive animations** with **mouse hover** and falling blooms.

Shattered Wonderland, Platformer Game

- Developed a **platformer-horror game** inspired by Alice in Wonderland, using **Unreal Engine** as a **Team Project** for Game Development Course.
- Implemented **mechanics** and features for the first level, using **C++ scripting** and **Blueprints**.

PawQuants

- Designed and Developed a **Top-Down Pixel Art** Game with **Quantum Gates** Changing Player's Movement Using **Unity 2D** playable in **browser**

Museumore , Augmented Reality Web Application For Museums and Galleries

- Built a responsive app frontend with **React**, consisting of **10** mobile-friendly pages designed in **Figma**.
- Developed backend with **Django**, utilized **Python image processing** for compiling image targets.
- Leveraged **mindAR** and **ThreeJS** libraries for AR development. Successfully diagnosed **over 100** images, particularly paintings.

Othello Game , An implementation of an AI player for the board game Othello

- Developed Othello game using **Java** and **JavaFX**, featuring AI with **Minimax algorithm** and **Alpha-Beta** pruning.

EDUCATION

Master of Science in Interactive Media, FH Univeristy of Upper Austria

Game Development Track

Focusing on immersive technologies and advanced rendering systems with key modules in **Visual Computing**, **Spatial Computing**, **Reinforcement Learning**, and **Real-time Engineering**.

2024 – present
Hagenberg, Austria

Bachelor of Science in Computer Engineering, Shahid Beheshti University

Strengthened vital programming skills through pivotal courses such as **Internet Engineering**, **Data Structures**, **Database Fundamentals**, **Algorithm Design**, **Artificial Intelligence** and more.

2018 – 2023
Tehran, Iran

LANGUAGES

Persian

Native

English

IELTS Academic 7.5, Proficient

Deutsch

A2 (Actively Learning)